Local Average and Marginal Treatment Effects: Identification, Estimation, and Monte Carlo Evidence

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Abstract

We examine the identification and estimation of local average treatment effects (LATE) and marginal treatment effects (MTE) in settings with a binary endogenous variable and one or more instrumental variables. We provide a unified treatment of the Imbens-Angrist (1994) LATE theorem and the Heckman-Vytlacil (2005) MTE framework, clarifying their relationship and showing that IV estimates are weighted averages of MTEs with instrument-specific weights. Monte Carlo evidence demonstrates that 2SLS estimators are approximately unbiased in samples exceeding 500 observations, but standard asymptotic confidence intervals under-cover by 4 percentage points at n = 200; Anderson-Rubin confidence sets achieve exact coverage at all sample sizes. In an empirical application using three randomly assigned encouragement instruments, we estimate a LATE of 1.36 that is robust across specifications and overidentification tests, and a downward-sloping MTE function consistent with diminishing returns to treatment at higher resistance levels. We derive sharp bounds on the ATE without monotonicity, finding that Machina-Storti IV bounds [0.78, 1.94] include the LATE.

Contents

1. Introduction

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We contribute to the literature on applied econometrics by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in applied econometrics since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

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2. Framework and Notation

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss

the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

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2.1. Potential Outcomes and Treatment Assignment

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$$Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}, \quad D_i = Z_i D_{1i} + (1 - Z_i) D_{0i} \tag{1}$$

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2.2. Population Heterogeneity and Types

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$$\text{LATE} = E[Y_{1i} - Y_{0i} \mid D_{1i} > D_{0i}] = \frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]} \tag{2}$$

$$\hat{\tau}_{Wald} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{D}_1 - \bar{D}_0} = \frac{\hat{\rho}_{YZ}}{\hat{\rho}_{DZ}} \tag{3}$$

$$D_{1i} \geq D_{0i} \quad \forall i \quad (\text{or} \quad D_{1i} \leq D_{0i} \quad \forall i) \tag{4}$$

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Table 1: Characteristics of Compliers, Always-Takers, and Never-Takers

	Complier Mean	Always-taker Mean	Never-taker Mean
Age	34.2	31.8	38.1
Female (%)	52.1	48.7	55.3
Years of schooling	12.4	14.2	10.8
Baseline income ($000s)	38.7	52.1	28.4
Prior program participation (%)	18.3	41.2	9.4
Share of population	0.40	0.22	0.38

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3.3. Bounds Without Monotonicity

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a

growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

$$E[Y_1 - Y_0] \in \left[\text{LATE} + \frac{p_d - p_0}{1 - p_0 + p_d}\underline{Y},\ \text{LATE} + \frac{p_d}{p_0 + 1 - p_d}\overline{Y}\right] \tag{5}$$

A central question in applied econometrics is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable

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Table 2: Bounds on the Average Treatment Effect Without Monotonicity

	Lower Bound	Point (LATE)	Upper Bound	Bound Width
Manski (1990) no-assumptions bounds	-8.41	1.36	8.41	16.82
Monotone treatment response	0.00	1.36	8.41	8.41
Monotone treatment selection	-1.84	1.36	4.12	5.96
MTR + MTS combined	0.00	1.36	4.12	4.12
Machina-Storti (IV bounds)	0.78	1.36	1.94	1.16
Horowitz-Manski (outcome support)	0.61	1.36	2.11	1.50

Notes: Bounds on the ATE for the full population. LATE is the 2SLS point estimate using all three instruments (1.36). MTR = monotone treatment response. MTS = monotone treatment selection. Machina-Storti bounds use the instruments as identifying information. Outcome Y is bounded in [0, 10].

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4. Marginal Treatment Effects

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$$\text{MTE}(x, u) = E[Y_{1i} - Y_{0i} \mid X_i = x, U_i = u] \tag{6}$$

$$\text{MTE}(x,u) = \left.\frac{\partial E[Y \mid X = x, P(Z) = p]}{\partial p}\right|_{p=u}, \quad \lfloor n \cdot u \rfloor \leq \text{rank}(p_i) \leq \lceil n \cdot u \rceil \tag{7}$$

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

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4.1. Local IV Estimation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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4.2. MTE and IV Weights

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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A central question in applied econometrics is whether changes in the treatment variable lead to

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$$E[Y_i(1) - Y_i(0)] = E\left[\frac{D_i Y_i}{P(Z_i)} - \frac{(1-D_i)Y_i}{1-P(Z_i)}\right] \tag{8}$$

$$\text{LATE} = \int_0^1 \text{MTE}(u)\, h_{LATE}(u)\, du, \quad h_{LATE}(u) = \frac{\mathbf{1}(u \le p_1) - \mathbf{1}(u \le p_0)}{p_1 - p_0}, \quad \hat{\tau} = \arg\max_{\tau} \mathcal{L}(\tau \mid \mathbf{Y}, \mathbf{D}, \mathbf{Z}) \tag{9}$$

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Table 3: Marginal Treatment Effects by Propensity Score Decile

	MTE	95% CI Low	95% CI High	IV Weight
Decile 1 (u = 0.05)	2.14***	1.41	2.87	0.031
Decile 2 (u = 0.15)	1.91***	1.31	2.51	0.067
Decile 3 (u = 0.25)	1.72***	1.21	2.23	0.108
Decile 4 (u = 0.35)	1.54***	1.08	2.00	0.141
Decile 5 (u = 0.45)	1.39***	0.97	1.81	0.162
Decile 6 (u = 0.55)	1.21***	0.81	1.61	0.158
Decile 7 (u = 0.65)	1.04***	0.62	1.46	0.138
Decile 8 (u = 0.75)	0.87***	0.41	1.33	0.101
Decile 9 (u = 0.85)	0.68***	0.14	1.22	0.072
Decile 10 (u = 0.95)	0.44*	-0.18	1.06	0.022

Notes: *** p<0.01, ** p<0.05, * p<0.1. MTE estimated via local IV (Heckman-Vytlacil method). IV Weight is the 2SLS weight on each MTE cell under the three instruments. Weights sum to 1.0.

4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

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4.3. Policy-Relevant Treatment Effects

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5. Monte Carlo Design

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Table 4: Appendix: Monte Carlo DGP Specifications

	DGP 1	DGP 2	DGP 3	DGP 4
True LATE	1.00	1.50	1.00	2.00
True ATE	1.20	1.50	0.80	2.00
Selection (rho)	0.50	0.50	0.70	0.00
Compliance rate	0.40	0.40	0.25	0.60
Instrument strength (E[Z-compliance])	0.40	0.40	0.25	0.60
Outcome variance	1.00	2.25	1.00	4.00

Notes: DGP 1 is the baseline; DGP 2 has a larger effect and more variance; DGP 3 has one-sided noncompliance (low compliance, strong selection); DGP 4 has no selection (rho=0, LATE=ATE).

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

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5.1. Data Generating Processes

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the

10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for

breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

5.2. Estimators Considered

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on applied econometrics by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in applied econometrics is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically

meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in applied econometrics since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

6. Results

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treat-

ment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the

Table 5: Monte Carlo: Consistency of LATE Estimators (N = 1,000 replications)

	n=200	n=500	n=1000	n=5000
Panel A: Bias (true LATE = 1.000)				
Wald / 2SLS	-0.042	-0.017	-0.009	-0.002
Control function (bilinear)	-0.038	-0.015	-0.007	-0.001
LIML	-0.011	-0.004	-0.002	0.001
OLS (biased benchmark)	0.312	0.318	0.321	0.319
Panel B: RMSE				
Wald / 2SLS	0.241	0.148	0.104	0.046
Control function (bilinear)	0.228	0.141	0.099	0.044
LIML	0.219	0.136	0.097	0.043
OLS (biased benchmark)	0.318	0.319	0.321	0.319

Notes: DGP: binary instrument, binary treatment, continuous outcome. Compliance rate = 0.40. Selection on unobservables: rho = 0.50. True LATE = 1.000 by construction. All IV estimators use the single binary instrument.

Table 6: Monte Carlo: Coverage Rates of 95% Confidence Intervals for LATE

	n=200	n=500	n=1000	n=5000
Panel A: Standard (asymptotic) confidence intervals				
2SLS (asymptotic SE)	0.906	0.928	0.938	0.947
2SLS (heteroskedastic-robust SE)	0.912	0.934	0.942	0.949
LIML (asymptotic SE)	0.931	0.941	0.946	0.950
Panel B: Bootstrap confidence intervals (999 replications)				
2SLS (percentile bootstrap)	0.921	0.938	0.944	0.950
2SLS (BC bootstrap)	0.934	0.944	0.948	0.951
Anderson-Rubin (weak-IV robust)	0.948	0.950	0.950	0.950

Notes: Coverage rates across 1,000 Monte Carlo replications. Nominal coverage = 0.95. DGP identical to Table **??**. BC = bias-corrected bootstrap. Anderson-Rubin CI is exact under the null regardless of instrument strength.

Table 7: Monte Carlo: MTE Estimation – RMSE and Bias by Propensity Score Percentile

	p10-p20	p30-p40	p50-p60	p70-p80	p80-p90
Panel A: Bias (true MTE by cell)					
Local IV (Heckman-Vytlacil)	-0.082	-0.041	-0.018	-0.037	-0.091
Polynomial series (degree 3)	-0.064	-0.029	-0.011	-0.028	-0.071
Local polynomial (bw = 0.15)	-0.071	-0.033	-0.014	-0.031	-0.078
Panel B: RMSE					
Local IV (Heckman-Vytlacil)	0.198	0.142	0.118	0.147	0.211
Polynomial series (degree 3)	0.167	0.121	0.101	0.127	0.182
Local polynomial (bw = 0.15)	0.183	0.133	0.110	0.138	0.196

Notes: n = 2,000. 500 replications. True MTE is decreasing in the propensity score: MTE(u) = 2.0 - 2.0u. Cells defined by propensity score percentile intervals. Boundary cells (p10-p20, p80-p90) exhibit larger bias due to boundary effects in local estimation.

first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

6.1. Finite-Sample Performance of LATE Estimators

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treat-

ment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C

restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

6.2. Coverage Properties

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on applied econometrics by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in applied econometrics is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in applied econometrics since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

6.3. MTE Estimation Results

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

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We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

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outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

7. Empirical Application

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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$$J = n \cdot \hat{g}'\hat{W}^{-1}\hat{g} \xrightarrow{d} \chi^2(L-K), \quad \hat{g} = \frac{1}{n}\sum_{i=1}^{n} Z_i(Y_i - D_i\hat{\tau}), \quad \hat{W} = \begin{bmatrix} \hat{\sigma}_{11} & \cdots & \hat{\sigma}_{1L} \\ \vdots & \ddots & \vdots \\ \hat{\sigma}_{L1} & \cdots & \hat{\sigma}_{LL} \end{bmatrix} \tag{10}$$

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve

the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Table 8: Wald Estimates by Instrument

	Instrument 1	Instrument 2	Instrument 3	Joint
Dep. var.: Outcome Y				
Wald estimate (LATE)	1.42***	1.31***	1.38***	1.36***
	((0.28))	((0.31))	((0.34))	((0.18))
First-stage coefficient	0.41***	0.35***	0.28***	
	((0.04))	((0.04))	((0.05))	
Reduced-form coefficient	0.58***	0.46***	0.39***	
	((0.11))	((0.12))	((0.14))	
First-stage F	104.2	76.8	31.4	
Observations	8,412	8,412	8,412	8,412

Notes: *** p<0.01, ** p<0.05, * p<0.1. Each column uses a single instrument to form a Wald estimate. Joint uses all three instruments in 2SLS. Instruments are randomly assigned encouragement doses in a field experiment.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to

Table 9: Two-Stage Least Squares Estimates

	(1)	(2)	(3)	(4)
Panel A: Second Stage				
Dep. var.: Outcome Y				
Treatment D	1.36***	1.31***	1.28***	1.24***
	((0.18))	((0.17))	((0.16))	((0.16))
Age		0.02**	0.02**	0.02**
		((0.01))	((0.01))	((0.01))
Female		-0.14***	-0.13***	-0.13***
		((0.04))	((0.04))	((0.04))
Panel B: First Stage				
Dep. var.: Treatment D				
Instrument Z1	0.41***	0.40***	0.39***	0.39***
	((0.04))	((0.04))	((0.04))	((0.04))
Instrument Z2	0.35***	0.34***	0.34***	0.33***
	((0.04))	((0.04))	((0.04))	((0.04))
Instrument Z3	0.28***	0.27***	0.27***	0.27***
	((0.05))	((0.05))	((0.05))	((0.05))
Demographic controls	No	Yes	Yes	Yes
Site FE	No	No	Yes	Yes
Cohort FE	No	No	No	Yes
Observations	8,412	8,412	8,412	8,412
First-stage F (joint)	104.2	98.7	91.3	89.4
Hansen J p-value	0.84	0.81	0.79	0.77

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered by site. Hansen J p-values indicate failure to reject the null of instrument validity.

Table 10: Specification Tests

	Statistic	p-value	Null hypothesis	Verdict
Hansen J (overid.)	0.47	0.79	Instruments valid	Fail to reject
Hausman (OLS vs. 2SLS)	18.4	0.00	OLS consistent	Reject
First-stage F (joint)	89.4	0.00	Instruments irrelevant	Reject
Stock-Yogo 10Monotonicity (Mourifie-Wan)	0.81	0.47	Monotonicity violated	Fail to reject
LATE = ATE (Vytlacil)	1.84	0.17	MTE constant	Fail to reject

Notes: Hausman test uses robust variance matrix. Stock-Yogo 10% maximal bias critical value is 13.9 for 3 instruments; F-stat of 89.4 comfortably exceeds this. Monotonicity test uses the nonparametric test of Mourifie and Wan (2017).

internal validity that have limited the conclusions of prior work.

We contribute to the literature on applied econometrics by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

7.1. Wald and 2SLS Estimates

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

7.2. Specification Tests

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on applied econometrics by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy,

which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in applied econometrics is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central

topic in applied econometrics since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

8. Extensions

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time

period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

8.1. Multiple Instruments and Instrument Validity

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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8.2. Heterogeneous Effects and Interaction Models

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key

source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the

underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

9. Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects

the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

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10. Appendix A: Proofs

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications

and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

We provide here a detailed description of the data sources and variable construction procedures. For each variable, we report the original source, the years available, the unit of observation, and any transformations applied. We also report summary statistics for the full set of variables used in the analysis, including those that appear only in robustness checks or auxiliary regressions.

The following tables report the complete set of coefficient estimates for all control variables included in our main specifications. In the interest of space, the main text reports only the coefficients on the key independent variables. The full results presented here confirm that the control variables enter with expected signs and magnitudes, and that their inclusion does not materially affect the coefficients of interest.

A.1 Proof of LATE Identification (Imbens-Angrist 1994)

Proposition 1 (LATE Identification)**.** *Under (i) independence $Z \perp (Y_0, Y_1, D_0, D_1)$, (ii) relevance $E[D_1 - D_0] \neq 0$, and (iii) monotonicity $D_{1i} \geq D_{0i}$ for all i, the Wald estimator identifies the LATE:*

$$\frac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{E[D \mid Z = 1] - E[D \mid Z = 0]} = E[Y_1 - Y_0 \mid D_1 > D_0] \tag{11}$$

Proof. By the law of iterated expectations and independence:

$$\begin{aligned} E[Y \mid Z = 1] &= E[D_1 Y_1 + (1 - D_1) Y_0] \\ &= E[Y_1 \mid D_1 = 1] P(D_1 = 1) + E[Y_0 \mid D_1 = 0] P(D_1 = 0) \end{aligned} \tag{12}$$

Similarly, $E[Y \mid Z = 0] = E[Y_1 \mid D_0 = 1] P(D_0 = 1) + E[Y_0 \mid D_0 = 0] P(D_0 = 0)$.

Taking the difference and using monotonicity to partition the population into always-takers

($D_0 = D_1 = 1$), compliers ($D_0 = 0, D_1 = 1$), and never-takers ($D_0 = D_1 = 0$):

$$\begin{aligned} E[Y \mid Z = 1] - E[Y \mid Z = 0] &= E[(Y_1 - Y_0)(D_1 - D_0)] \\ &= E[Y_1 - Y_0 \mid D_1 > D_0] \cdot P(D_1 > D_0) \end{aligned} \tag{13}$$

Dividing by $E[D \mid Z = 1] - E[D \mid Z = 0] = E[D_1 - D_0] = P(D_1 > D_0)$ completes the proof. □

A.2 MTE as the Limit of LATE

Let Z be continuously distributed and $p = P(D = 1 \mid Z = z) = p(z)$. Consider a pair of values z_0, z_1 with $p_0 = p(z_0) < p_1 = p(z_1)$. The Wald estimator identifies:

$$\frac{E[Y \mid P(Z) = p_1] - E[Y \mid P(Z) = p_0]}{p_1 - p_0} = E[Y_1 - Y_0 \mid p_0 < U_D \leq p_1] \tag{14}$$

where U_D is the rank of the individual in the selection equation. Taking the limit $p_1 \to p_0$:

$$\lim_{p_1 \to p_0} \frac{E[Y \mid P(Z) = p_1] - E[Y \mid P(Z) = p_0]}{p_1 - p_0} = \left. \frac{\partial E[Y \mid P(Z) = p]}{\partial p} \right|_{p=p_0} = \text{MTE}(p_0) \tag{15}$$

This establishes Equation **??**.

A.3 IV Weights Integrate to One

Lemma 1 (Unit Sum of IV Weights)**.** *For a binary instrument $Z \in \{0, 1\}$, the IV weights $h_{IV}(u) \propto F_1(u) - F_0(u)$ satisfy $\int_0^1 h_{IV}(u)\, du = 1$.*

Proof.

$$\begin{aligned} \int_0^1 h_{IV}(u)\, du &= \frac{\int_0^1 (F_1(u) - F_0(u))\, du}{E[D|Z = 1] - E[D|Z = 0]} \\ &= \frac{E[D|Z = 1] - E[D|Z = 0]}{E[D|Z = 1] - E[D|Z = 0]} = 1 \end{aligned} \tag{16}$$

where we used $\int_0^1 F_j(u)\, du = E[D \mid Z = j]$ for $j \in \{0, 1\}$ by the fundamental theorem of calculus applied to the CDF of U_D. □

A.4 Policy-Relevant Treatment Effects

For a policy that changes the propensity score from p to $p' = p+\delta$, the policy-relevant treatment effect is:

$$\text{PRTE}(\delta) = \frac{1}{\delta}\int_{p}^{p+\delta} \text{MTE}(u)\,du \approx \text{MTE}(p) \quad \text{for small } \delta \tag{17}$$

This shows that for marginal policies (small δ), the relevant parameter is the MTE at the margin, not the ATE or LATE.

A.5 Matrix Representation of 2SLS with Multiple Instruments

The 2SLS estimator using instruments $\mathbf{Z} \in \mathbb{R}^{n\times L}$ can be written in matrix form:

$$\hat{\tau}_{2SLS} = \arg\min_{\tau} \ \|\mathbf{P}_Z(\mathbf{Y} - \mathbf{D}\tau)\|^2, \quad \mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}' = \begin{bmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{bmatrix}. \tag{18}$$

A.6 Floor/Ceiling Bin Construction for Local Estimation

The local polynomial MTE estimator uses propensity score cells defined by $\lfloor n\cdot u_L \rfloor \leq \text{rank}(p_i) \leq \lceil n \cdot u_H \rceil$, where the bandwidth h determines $u_L = u - h$ and $u_H = u + h$.

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

11. Appendix B: MTE Derivations

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

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12. Appendix C: Monte Carlo Details

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